Sergei Volkov

Senior Data & MLOps Engineer

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Work Authorization: EU Blue Card holder (>2 yrs) · Unrestricted German Work Permit (incl. Freiberufler / Commercial)

Senior Data & MLOps Engineer and active NixOS package maintainer with 6+ years designing and scaling production data platforms across gaming analytics (Playrix, 50M+ DAU), enterprise e-commerce (Zalando), and ML experimentation (AppIQ). Specialized in self-hosted lakehouses (DuckDB/DuckLake on AWS S3, Delta Lake), orchestration (Apache Airflow, CI/CD), and low-latency ML model serving. Official package maintainer in `nixpkgs` (Apache Airflow, Julia, Databricks CLI). Co-author of 5 peer-reviewed publications in computational fluid dynamics.

WORK EXPERIENCE

Freelance Data / MLOps Engineer | AppIQ
Jan 2026 – Present

Architecture & Implementation of ML-driven Experimentation Platform

- Architected and deployed a self-hosted Lakehouse (**DuckDB/DuckLake on AWS S3**) via **Terraform**, delivering sub-second analytical query latency while eliminating Spark operational overhead.
- Built low-latency ML model serving infrastructure and automated training/evaluation pipelines for real-time model inference.
- Designed high-throughput event ingestion and user tracking pipelines, enabling automated A/B test assignment and real-time metric calculation.
- Standardized declarative infrastructure-as-code and reproducible deployment workflows using **Terraform** and **Nix**.

Sabbatical & Open Source Engineering | Self-Directed
Apr 2025 – Sep 2025

Open Source Contributions & Advanced Technical Upskilling

- Dedicated sabbatical for intensive open-source contributions to **NixOS** (`nixpkgs`) and packages' upstream, packaging and maintaining **Apache Airflow**, **Julia**, and **Databricks CLI**.
- Conducted architectural research on embedded analytical engines (DuckDB) and prepared for advanced machine learning training.

Data Engineer | Zalando SE
Oct 2024 – Mar 2025

Central Data Platform & Analytics Engineering Services

- Developed and maintained business-critical ETL/ELT pipelines using **Managed Workflows for Apache Airflow (MWAA)** and **Databricks Delta Lake**, supporting 15+ internal stakeholder teams.
- Engineered automated CI/CD deployment pipelines for internal Python helper packages and notebook workloads across staging and production Databricks environments.
- Conducted fraudulent customers detection dataset design.
- Provided code reviews, data QA, and architectural support for cross-functional data consumers.

Data Scientist & Cloud Architect | Crocus Labs GmbH
Aug 2023 – Feb 2024

IoT Platform Architecture & Computational Photometry

- Architected scalable cloud infrastructure on **AWS (IoT Core, Lambda, S3, DynamoDB)** for real-time telemetry ingestion from smart greenhouse LED hardware.
- Developed a Python raytracing simulation to model 3D light distribution and optimize LED chip array layouts based on raw photometric emitter data.
- Engineered a containerized IoT test harness with virtualized sensor clients and an **MQTT broker**, enabling end-to-end integration testing without physical hardware.

Data Quality Engineer / Analytics Engineer | **Playrix** Nov 2019 – Feb 2023
Promoted across 3 roles (Data Analyst -> QA Engineer -> Data Quality Engineer) for 50M+ DAU platforms

- Led data quality assurance during the migration of legacy Big Data pipelines to a modern lakehouse architecture across **Spark, Hive, Redshift, and Snowflake**.
- Built dynamic SQL metaprogramming validation frameworks that automated data anomaly detection across billions of daily in-game event records.
- Designed and optimized executive **Tableau** dashboards and underlying dimensional data marts tracking player retention, monetization, and A/B test experiments.
- Managed a QA sub-team, establishing automated regression test suites and data verification protocols for engineering releases.

Research Engineer | **Leibniz Institute of Freshwater Ecology (IGB Berlin)** Jun 2021 – Mar 2023
Computer Vision & Environmental Data Infrastructure

- Architected containerized computational environments (**Docker, CMake, MATLAB, OpenCV**) to modernize and execute a legacy 3D particle-tracking codebase.
- Developed computer vision data processing pipelines in **Python** to extract 3D velocity vectors from underwater stereo-video recordings of particulate biomass.
- Administered remote Linux compute nodes and automated data extraction workflows for gigabyte-scale video datasets.

Research Data Analyst | **Northern Water Problems Institute (NWPI)** Jan 2016 – Nov 2024
Telemetry Processing & Numerical Modeling

- Co-authored 5 peer-reviewed papers on convective turbulence and thermal stratification; processed long-term ADCP and thermistor telemetry.
- Built numerical analysis and data processing frameworks in **Python** and **MATLAB** for long-term limnological time-series data.

TECHNICAL COMPETENCIES

Data & Lakehouse:	Spark (PySpark), Databricks, DuckDB / DuckLake, Snowflake, BigQuery, dbt, Delta Lake, Redshift
Orchestration & Infra:	Apache Airflow (**NixOS Maintainer**), Docker, Podman, Kubernetes, Terraform, Nix/NixOS, CI/CD
Cloud & Storage:	AWS (S3, Lambda, IoT Core, MWAA, DynamoDB), GCP, PostgreSQL, MySQL
ML & Analytics:	Scikit-Learn, PyTorch, TensorFlow, MLflow, Model Serving, A/B Testing, Time-Series Analysis
Languages:	Python (NumPy, Pandas, SciPy), SQL, Bash, Julia, MATLAB

EDUCATION, QUALIFICATIONS & COMMUNITY

Data Science & Machine Learning Bootcamp Oct 2025 – Dec 2025
Ironhack

Specialist Degree in Energy Supplies (M.Sc. Equivalent) 2010 – 2015
Petrozavodsk State University

Machine Learning Specialization 2017
Stanford University on Coursera

Open Source: Official Package Maintainer for NixOS (`nixpkgs`: Apache Airflow, Julia, Databricks CLI, plus 6 Python libraries, system utilities) · OpenStreetMap Contributor
Languages: English (C1/C2 Full Professional), German (B2 Professional Working), Russian (Native)
Complete bibliography of 5 peer-reviewed publications and academic projects available in attached addendum or upon request.